



UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Data Structure and Algorithms II Lab (CSE 2218)
Summer – 2026 | Section F, P, Q, S
Assignment-1

Question 01: Longest Nice Substring:

10

A string *s* is **nice** if, for every letter of the alphabet that *s* contains, it appears **both** in uppercase and lowercase. For example, "abABB" is nice because 'A' and 'a' appear, and 'B' and 'b' appear. However, "abA" is not because 'b' appears, but 'B' does not.

Given a string *s*, return the longest **substring** of *s* that is **nice**. If there are multiple, return the substring of the **earliest** occurrence. If there are none, return an empty string.

Test Cases	Input	Output	Explanation
1	<i>s</i> = "YazaAay"	"aAa"	"aAa" is a nice string because 'A/a' is the only letter of the alphabet in <i>s</i> , and both 'A' and 'a' appear. "aAa" is the longest nice substring.
2	<i>s</i> = "Bb"	"Bb"	"Bb" is a nice string because both 'B' and 'b' appear. The whole string is a substring.
3	<i>s</i> = "c"	" "	There are no nice substrings.

Question 02: Find the Minimum SubArray using Divide and Conquer algorithm. 10

Given an array of integers, find the minimum possible sum of any contiguous subarray using the **divide and conquer** approach. A subarray must consist of consecutive elements from the array. Return the **minimum subarray sum**.

Test Cases	Input	Output
1	<i>arr[]</i> = {3, -4, 2, -3, -1, 7, -5}	-6
2	<i>arr[]</i> = {2, 6, 8, 1, 4}	1